Week 9
Lecture 11 (Chap 9)
Qualitative Data Analysis

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Summary of Data Gathering Types

	Raw Data	Example Qualitative Data	Example Quantitative Data	Initial Processing Steps
Interviews	Audio recordings. Interviewer notes. Video recordings.	Responses to open-ended questions. Video pictures. Respondent's opinions.	Age, number of mobile devices owned, years of experience. Responses to closed-ended questions.	Transcription of recordings. Expansion of notes. Entry of answers to closed-ended questions into a spreadsheet.
Questionnaires	Participant responses. Online database.	Responses to open-ended questions. Responses in “further comments” fields. Respondent's opinions.	Age, hours a week spent online, years of experience. Responses to closed-ended questions.	Clean up data. Filter into different datasets.
Observation	Observer's notes. Photographs. Audio and video recordings. Data logs. Think-aloud. Diaries.	Records of behavior. Description of a task as it is undertaken. Copies of documents outlining procedures.	Demographics of participants. Time spent on a task. The number of people involved in an activity. How many different types of activity are undertaken.	Expansion of notes. Transcription of recordings. Synchronization between data recordings.

Quantitative vs. Qualitative

Quantitative data is in the form of numbers, or data that can easily be translated into numbers. Examples are the number of years' experience the interviewees have, the number of projects a department handles at a time, or the number of minutes it takes to perform a task.

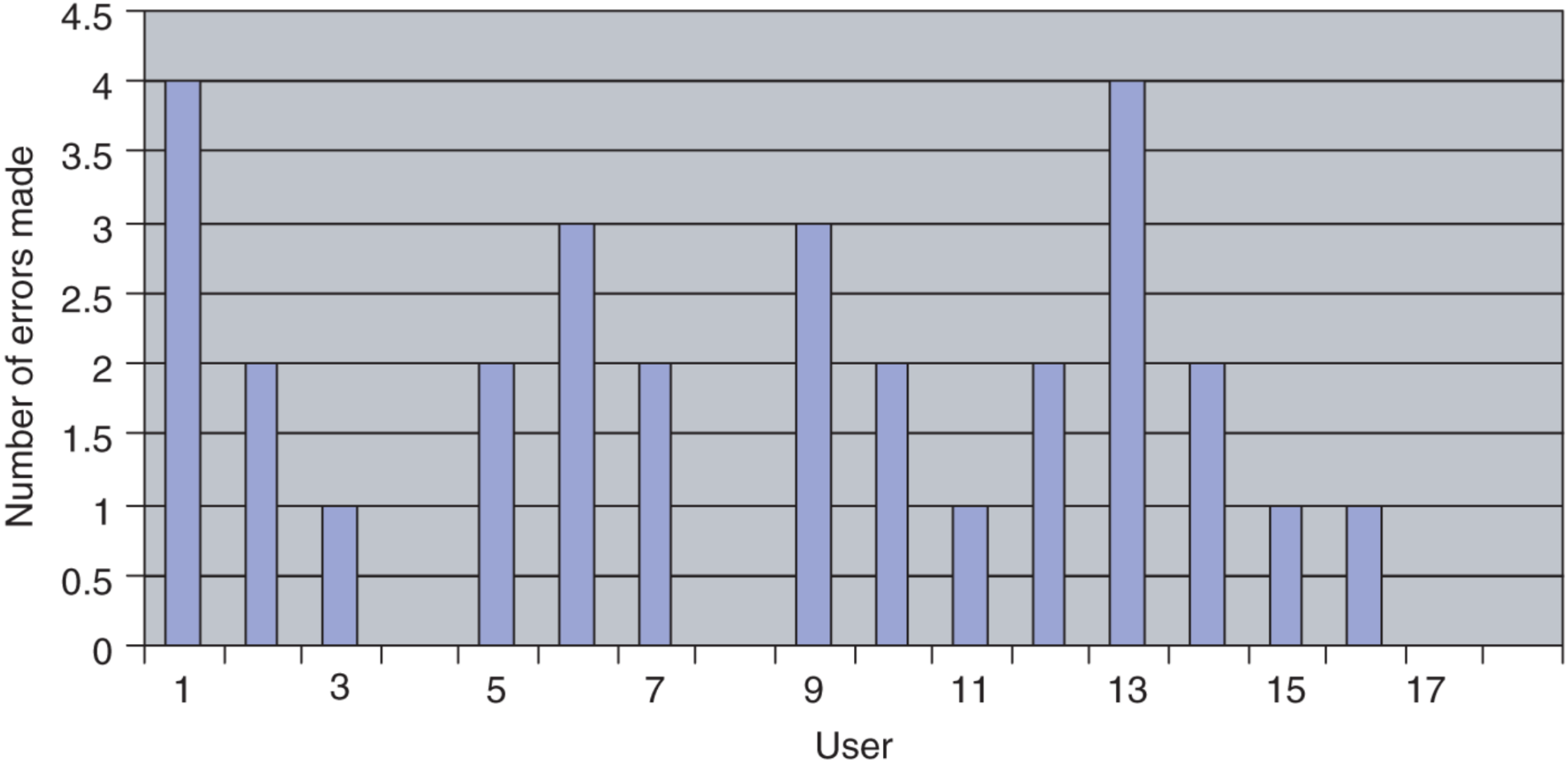
Qualitative data is in the form of words and images, and it includes descriptions, quotes from interviewees, vignettes of activity, and photos. It is possible to express qualitative data in numerical form, but it is not always meaningful to do so

Quantitative Data: Example (1)

Table 9.2 Data Gathered During a Study of a Photo Sharing App

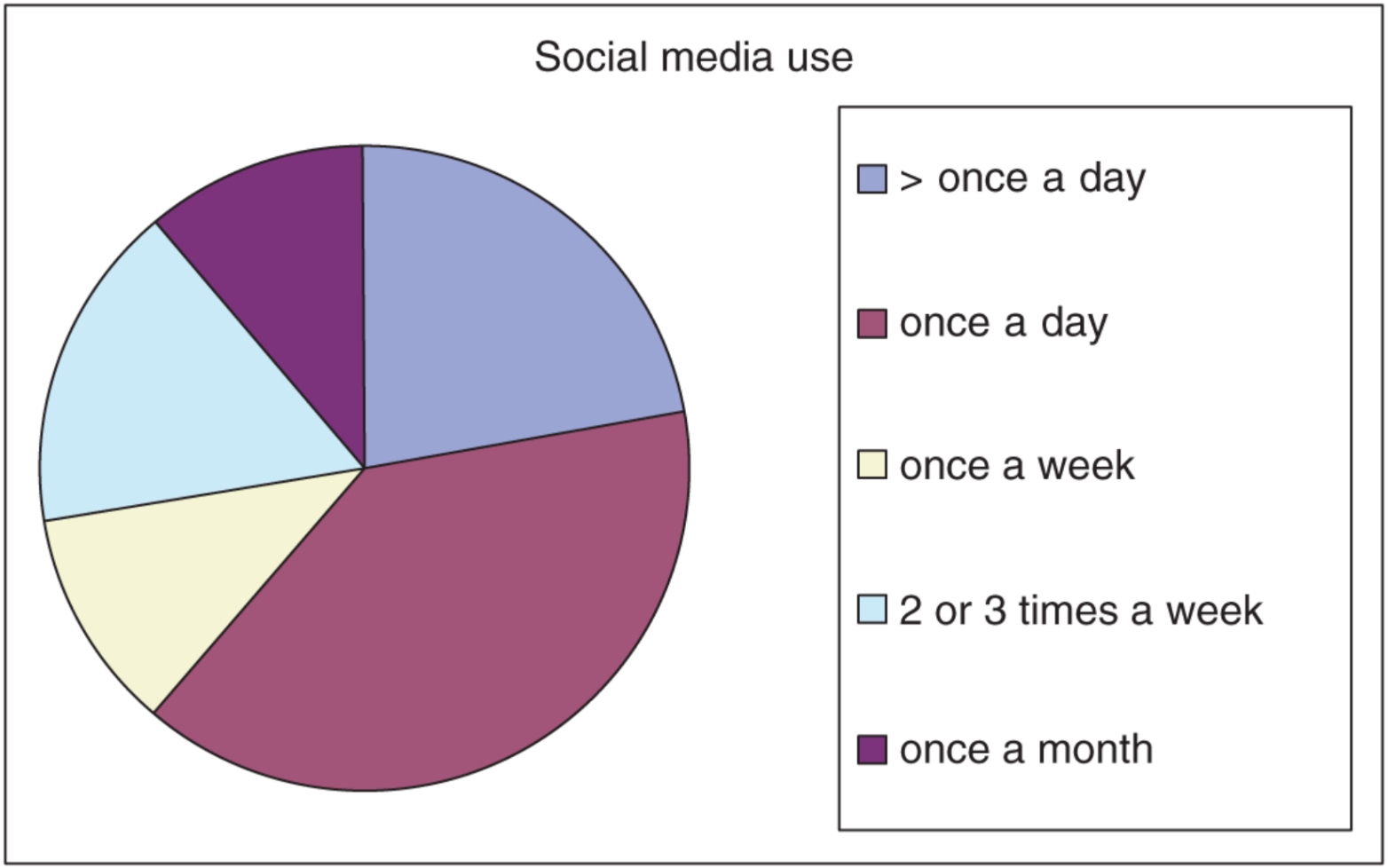
	Social Media Use					
User	**More Than Once a Day**	**Once a Day**	**Once a Week**	**Two or Three Times a Week**	**Once a Month**	**Number of Errors Made**
1		1				4
2	1					2
3			1			1
4	1					0
5				1		2
6		1				3
7	1					2
8		1				0
9					1	3
10	1					2
11				1		1
12			1			2
13		1				4
14		1				2
15					1	1
16				1		1
17		1				0
18		1				0
Totals	4	7	2	3	2	30
					Mean	**1.67**
					(to 2 decimal places)	

Quantitative Data: Example (2)

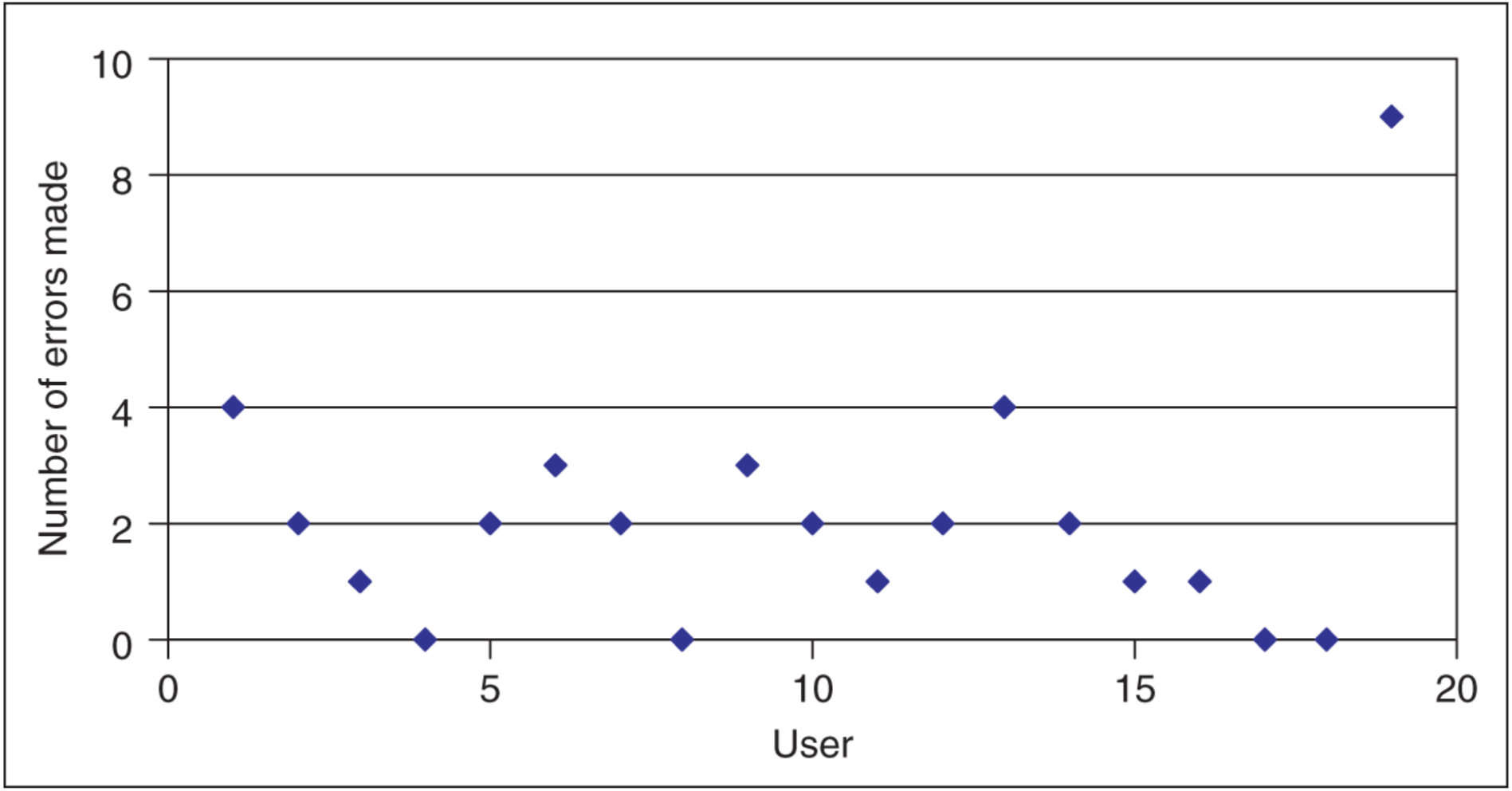


Quantitative Data: Example (3)

[R programming in one hour - a crash course for beginners - YouTube]



Quantitative Data: Example (4)



Systematic Qualitative Data Analysis (1)

- Qualitative data includes **attitudinal** data (thoughts, beliefs, needs) and **behavioral** data (observations of behavior)
- Analyzing data can be **overwhelming** due to large quantity of Information
- It is easy to be **biased by personal views** and **overlook** important insights
- Qualitative analysis can be systematic and **replicable**
- Different methods answer different questions

Systematic Qualitative Data Analysis (2)

Analyzing Video Material

A good way to start a video analysis is to watch what has been recorded all the way through while writing a high-level narrative of what happens, noting where in the video there are any potentially interesting events. How to decide which is an interesting event will depend on what is being observed. For example, in a study of the interruptions that occur in an open plan office, an event might be each time that a person takes a break from an ongoing activity, for instance, when a phone rings, someone walks into their cubicle, or email arrives. If it is a study of how pairs of students use an online collaborative learning tool, then activities such as turn-taking, speaking over one another, any exchanges in the chat, and periods when one or the other is distracted would be appropriate to record.

Chronological and video times can be used to index events. These may not be the same, since recordings can run at different speeds from real time and video can be edited. Video can be augmented with captured screens or logged data of people's interactions with a product, and transcription. There are various logging and screen capture tools available for this purpose, which enable interactions to be played back as a movie, showing screen objects being opened, moved, selected, and so on. These can then be played in parallel with the video to provide different perspectives on the talk, physical interactions, and the system's responses that occur. Having a combination of data streams can enable more detailed and fine-grained patterns of behavior to be interpreted (Heath et al., 2010).

Coding data using a coding scheme is integral to the analytic process. Anne Nassauer and Nicolas Legewie (2022) emphasize reliability as a guiding principle in analysis, meaning that the study is consistent in its procedure so that others can assess its research steps. They also emphasize the importance of considering alternative interpretations and provide some useful practical tips on video analysis, including how to construct clear and transparent concepts that enable reflection and critique, how to construct and apply a coding scheme, and how to overcome a range of challenges. For example, they highlight that coding decisions can be ambiguous, subjective, or incorrect. Even the best coding schemes can include codes that just don't work unambiguously all of the time. Subjectivity leads to different coding decisions by different researchers, and coding decisions may be incorrect simply because a mistake was made. To help overcome these challenges, they suggest making coding a team effort and drawing in expert knowledge of the study area.

Systematic Qualitative Data Analysis: Data Capture and Transcription

Proper data capture is crucial for successful analysis. This includes making detailed notes, transcribing and editing interviews, labeling media files, and breaking up large data sets into smaller, more manageable pieces.

Filtering data is necessary to focus on the most relevant and important information within the constraints of a research project.

Coding data

- Helps organize and make sense of it by identifying patterns and themes.
- Ongoing data processing and **debriefing with team** members can help ensure accuracy and identify emerging insights.

Thematic Analysis (TA): An Overview

- Thematic analysis is a method of organizing and analyzing qualitative data
- It involves finding themes by tagging observations and quotations with appropriate codes
- A theme is a description of a belief, practice, need, or other phenomenon discovered from the data
- Themes emerge when related findings appear multiple times across participants or data sources

TA: Challenges of Analyzing Qualitative Data

- **Large quantity of data** can be **time-consuming** to read and analyze
- Rich data with many details can lead to **superficial or descriptive analysis**
- **Contradicting data** can result in findings that are not definitive
- **Without clear goals**, analysis can lack focus and be misdirected
- Thematic analysis helps to **address these challenges** by providing a **systematic process** for organizing and analyzing data.

TA: Example showing diverse discussions related to addiction and recovery occurring on the subreddits (1)

Theme	Sub-Themes
Sharing Experiences	Self Reflections
	Sharing Failures Sharing Successes
	Waking Up
Peer Support	Check-ins
	Encouragement
	Solidarity
Consequences	Benefits of Recovery Costs of Recovery
	Harm from Substance Use
Substance Related Concerns	Pain Management
	Socializing

TA: Example showing diverse discussions related to addiction and recovery occurring on the subreddits (2)

Theme	Sub-Themes
Sharing Experiences	Self Reflections
	Sharing Failures Sharing Successes
	Waking Up
Peer Support	Check-ins
	Encouragement
	Solidarity
Consequences	Benefits of Recovery Costs of Recovery
	Harm from Substance Use
Substance Related Concerns	Pain Management
	Socializing

Paraphrased Example Quote

"Day 5, Monday Night. Really wanted to drink but I resisted!"
"Exercise works wonders! Try different activities like yoga and working out. Keep up the good effort!"
"It's great how our lives don't have to be like that anymore!"

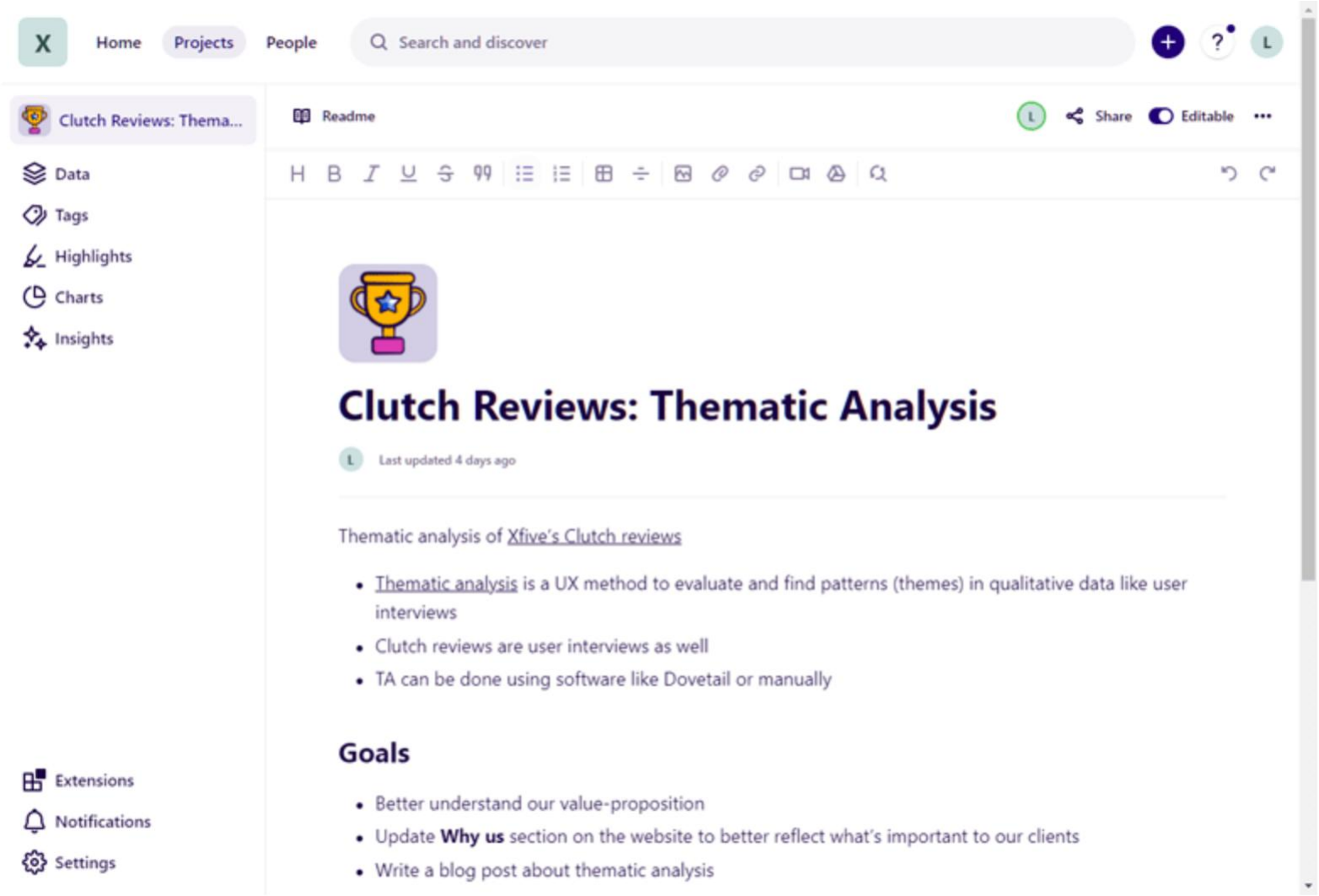
TA: Tools and Methods

- Thematic analysis can be done in various ways based on data, context, and researcher's personal style
- 3 common methods include: software, journaling, and affinity diagramming techniques

TA Using Software

- Data-analysis software can be used to code and analyze large amounts of qualitative data
- Examples of data-analysis software include Dovetail, EnjoyHQ, Delve, Aurelius, NVivo, Dedoose, and MAXQDA
- Benefits of using software include thorough analysis, ability to collaborate, and accessibility of raw data and results
- Drawbacks include time-consuming analysis, expense of subscriptions or licenses, and need for software learning

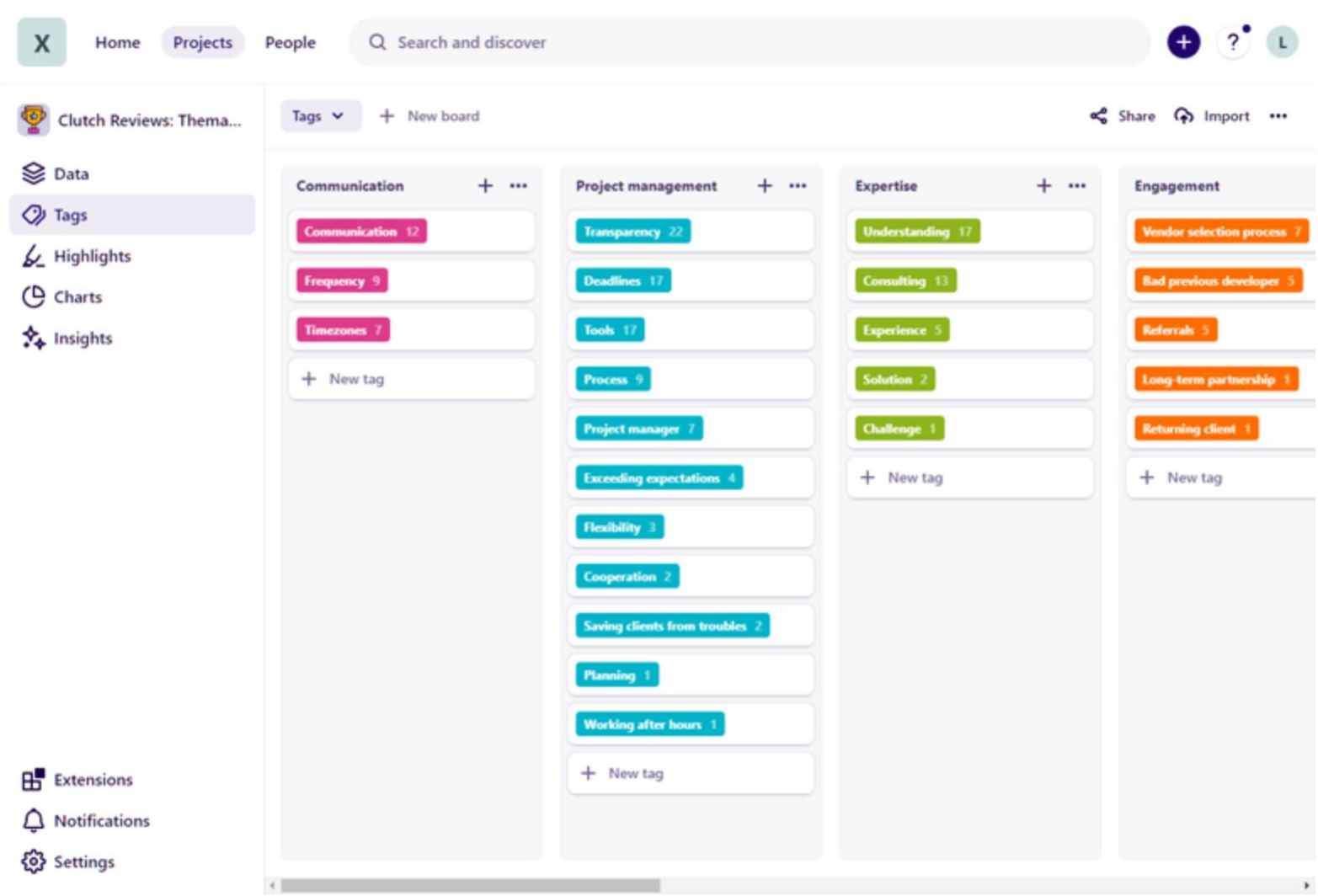
TA Using Software: Example (1)



Project page in Dovetail

https://www.xfive.co/blog/dovetail-thematic-analysis-clutch-reviews/

TA Using Software: Example (2)



https://www.xfive.co/blog/dovetail-thematic-analysis-clutch-reviews/

TA through Journaling

- Journaling involves manually annotating and highlighting data, followed by writing down thoughts and ideas
- This method encourages reflection and results in a record of how themes were arrived at
- Benefits include being cheap and flexible, but it can be difficult to do collaboratively

TA: Affinity Diagramming Techniques

- Affinity diagramming involves highlighting, cutting out, and reassembling data into meaningful groups until themes emerge
- Benefits include being able to be done collaboratively, quick theme identification, being cheap and flexible, and being visual and iterative
- Drawbacks include not being as thorough as other methods and being difficult to do with very varied or large amounts of data

Affinity Diagramming: Example (1)

https://www.nngroup.com/articles/affinity-diagram/

Affinity Diagramming: Example (2)

TA: Steps to Conduct

- Thematic analysis involves 5 phases: gathering data, reading all data, coding text, taking a break, and evaluating themes
- Phase 1: Gather all data, including transcripts, field notes, and diary study entries
- Phase 2: Read all data from beginning to end, and involve team in a workshop to engage with text and discuss findings
- Phase 3: Code text based on content, using descriptive codes and potentially interpretive codes
- Phase 4: Take a break and come back to the analysis later
- Phase 5: Evaluate themes for good fit, potentially involving team in review and discussion of themes

TA Phase 1: Gather Data

- Start with raw data, such as transcripts, field notes, and diary study entries
- *Transcribe audio recordings for analysis instead of relying on memory or notes*

A workshop where each team member reads each diary- or field-study entry and highlights important bits is a good way of getting team members to actively engage with the text, as opposed to just reading it and letting it wash over them.

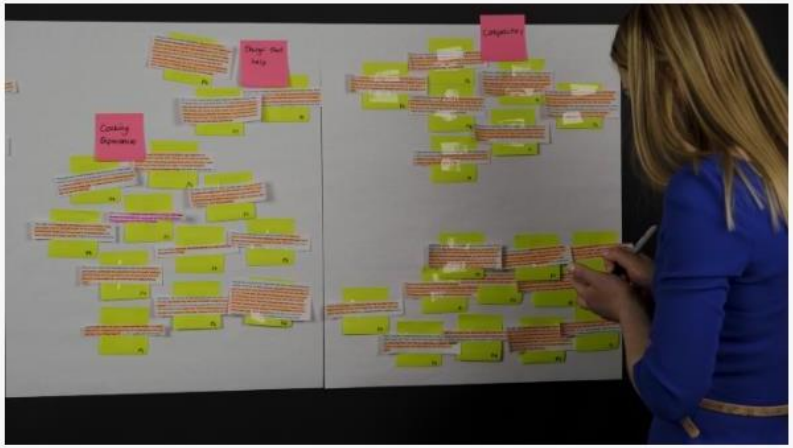
Source: NNGroup

TA Phase 2: Read All Data from Beginning to End

- **Familiarize** yourself with the data before beginning analysis
- Involve team in workshop to **engage** with text and discuss findings
- Print and display **research questions** for reference
- *Have each team member read and **highlight** important bits in transcripts or entries*
- **Discuss** as a group what was noticed or found surprising

Phase 3: Code Text Based on Content

- Review each segment of text and assign a descriptive code that describes the data
- Add interpretive codes if appropriate
- *Code can be created before or after grouping data*



The highlighted segments were clustered by the text topic and given a descriptive code.

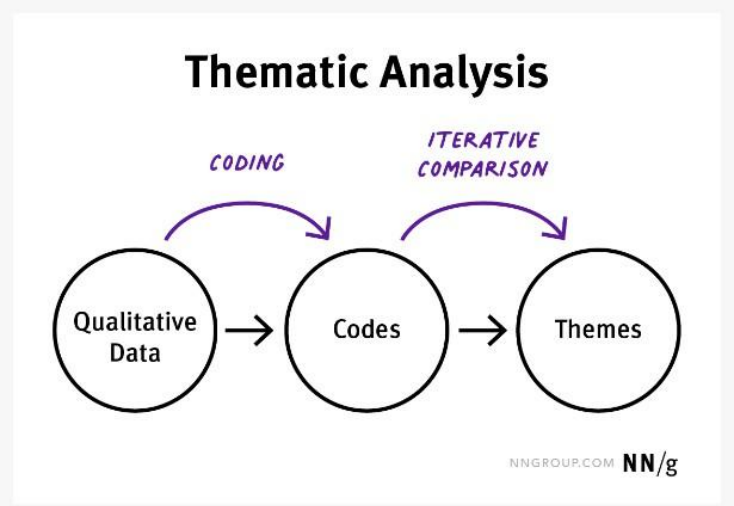
Source: NNGroup

TA: Introduction to Codes and Coding

- Definition: A code is a **label** for a **segment of text** that describes its content
- Good analogy: A code is like a keyword that describes an article or a hashtag that describes a tweet

TA: Introduction to Codes and Coding

- Coding: refers to the process of labeling text with codes
- Allows for easy comparison and analysis of data to uncover themes
- Often, researchers will also have a description and examples of each code to ensure consistency in coding



A thematic analysis starts with coding qualitative data. Through a systematic process of comparing segments of text within and between codes, the researcher arrives at themes.

Source: NNGroup

TA: Descriptive and Interpretive Codes

- Descriptive: They describe what the data is about

- Interpretive:

  They are an analytical reading of the data, adding the researcher's interpretive lens to it.

TA: Interview 1(1)

- Interviewer: Hello, can you please introduce yourself?
- Participant 1 (P1): Hi, I'm Sarah.

- Interviewer: Great, thank you for participating in this study. Can you tell us how often do you use Netflix?
- P1: I use Netflix almost every day.

TA: Interview 1(2)

- Interviewer: Hello, can you please introduce yourself?
- Participant 1 (P1): Hi, I'm Sarah.

- Interviewer: Great, thank you for participating in this study. Can you tell us how often do you use Netflix?
- P1: I use Netflix almost every day. [P1_DC_ daily user/active user]
  - P1: Participant 1
  - DC: Descriptive Code

- Frequency of Use: Participants use Netflix frequently, as seen in the quote "I use Netflix almost every day." (Frequency: 1)

TA: Interview 1(3)

- Interviewer: Can you describe your first experience with Netflix?
- P1: My first experience with Netflix was really good. I had heard a lot of good things about it and was excited to try it out. I signed up for the free trial and was really impressed with the selection of shows and movies available. I was hooked from the start.

TA: Interview 1(4)

- Interviewer: Can you describe your first experience with Netflix?
- P1: My first experience with Netflix was really good. I had heard a lot of good things about it and was excited to try it out [P1_DC_positive word of mouth]. I signed up for the free trial[P1_DC_free trial] and was really impressed with the selection of shows and movies available trial [P1_DC_ a lot of content]. I was hooked from the start.

- [P1_IC_positive first experience]
- [P1_IC_addictive first experience]
  - IC: Interpretive Code

TA: Interview 1(5)

Theme	Interpretive Code	Descriptive Code
Theme I: Positive First Experience: Participants had a positive first experience with Netflix, as seen in the quote/quotes "My first experience with Netflix was really good." (Frequency: 1)	[P1_IC_positive first experience] "My first experience with Netflix was really good."	[P1_DC_positive word of mouth]
	[P1_IC_addictive first experience] "I was hooked from the start."	[P1_DC_free trial]
		[P1_DC_ a lot of content]

- Frequency Count: The number of times an observation occurs (frequency) for a data item (variable) can be shown for both quantitative and qualitative data.
- For qualitative data this directly correlates with quotes, citations, references, snippets, etc.

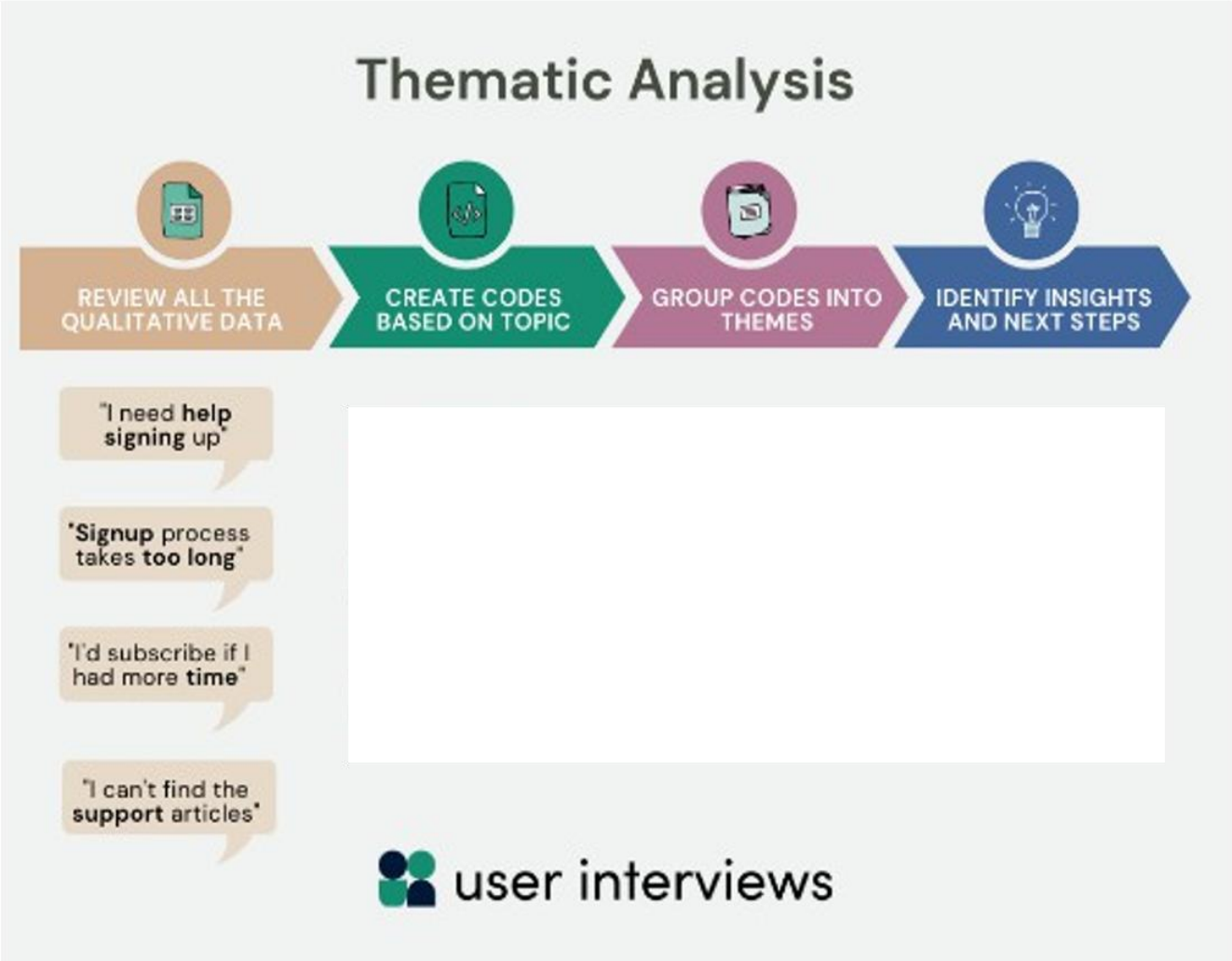
TA: Interview 1(6)

Theme	Interpretive Code	Descriptive Code
Theme I: Positive First Experience: Participants had a positive first experience with Netflix, as seen in the quote "My first experience with Netflix was really good." (Frequency: 1)	[P1_IC_positive first experience] "My first experience with Netflix was really good."	[P1_DC_positive word of mouth]
	[P1_IC_addictive first experience] "I was hooked from the start."	[P1_DC_free trial]
		[P1_DC_ a lot of content]

TA: Interview 1(7)

- Themes: Content Selection: Participants were impressed with the selection of shows and movies available on Netflix, as seen in the quote "I was really impressed with the selection of shows and movies available." (Frequency: 1)
  - Descriptive Code: NewContent
  - Interpretive Code: 'Always something new to watch'

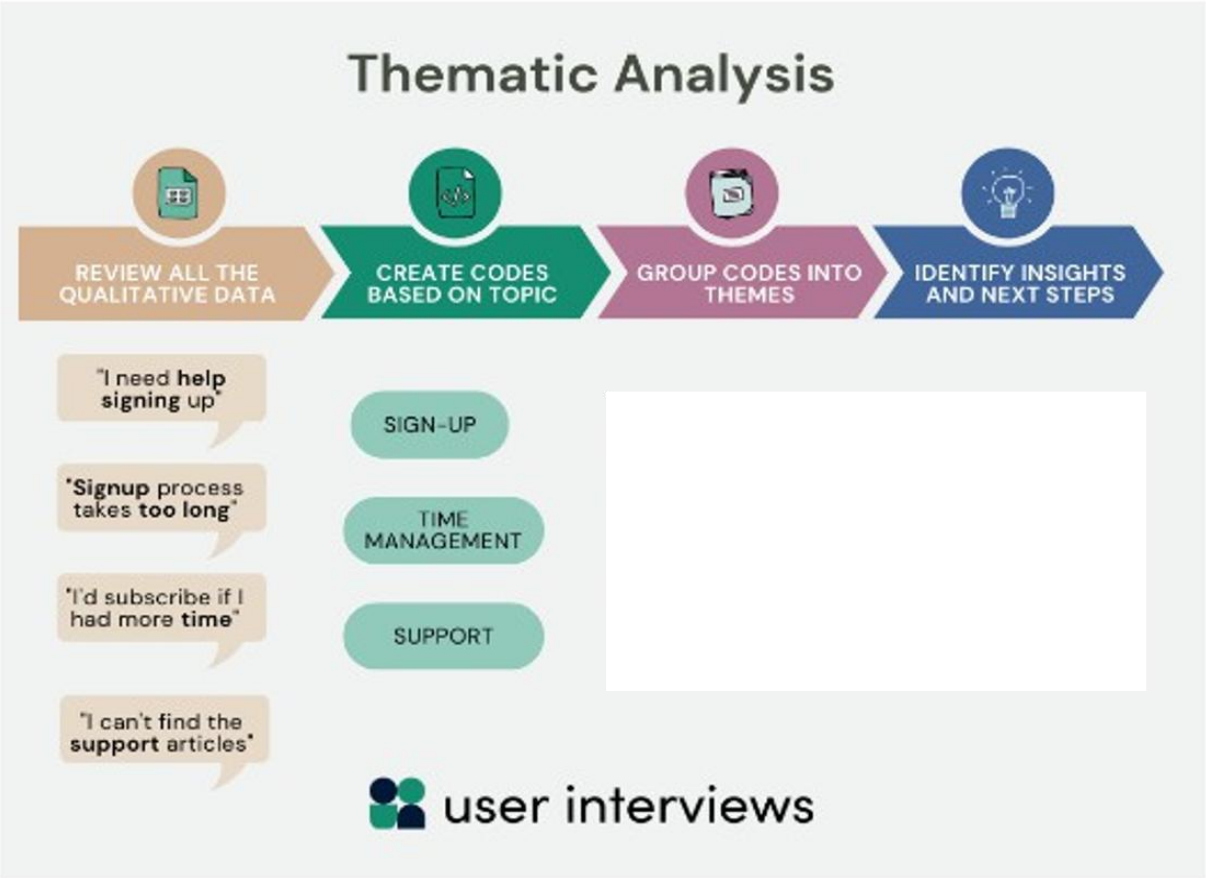
TA Process: Example (1)



Thematic analysis: Process and examples

Source: https://www.userinterviews.com/ux-research-field-guide-chapter/research-analysis

TA Process: Example (2)



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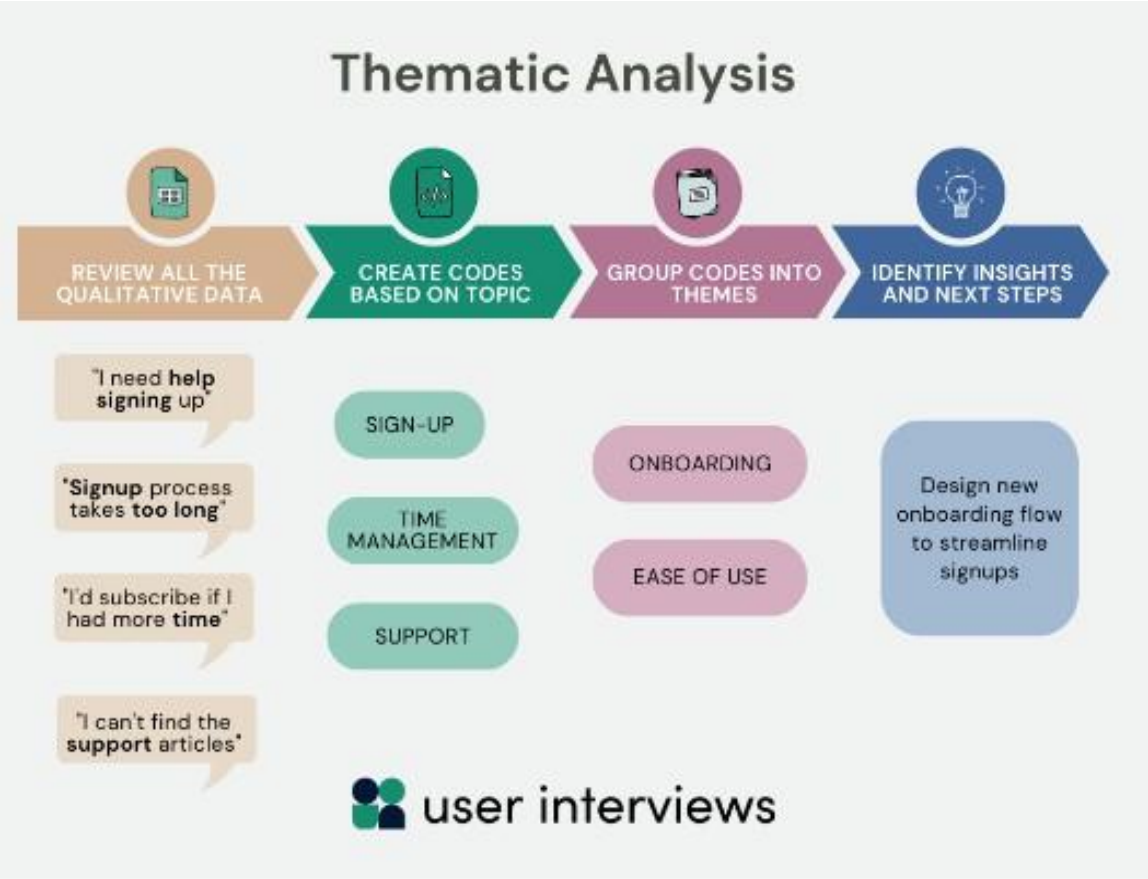
TA Process: Example (3)



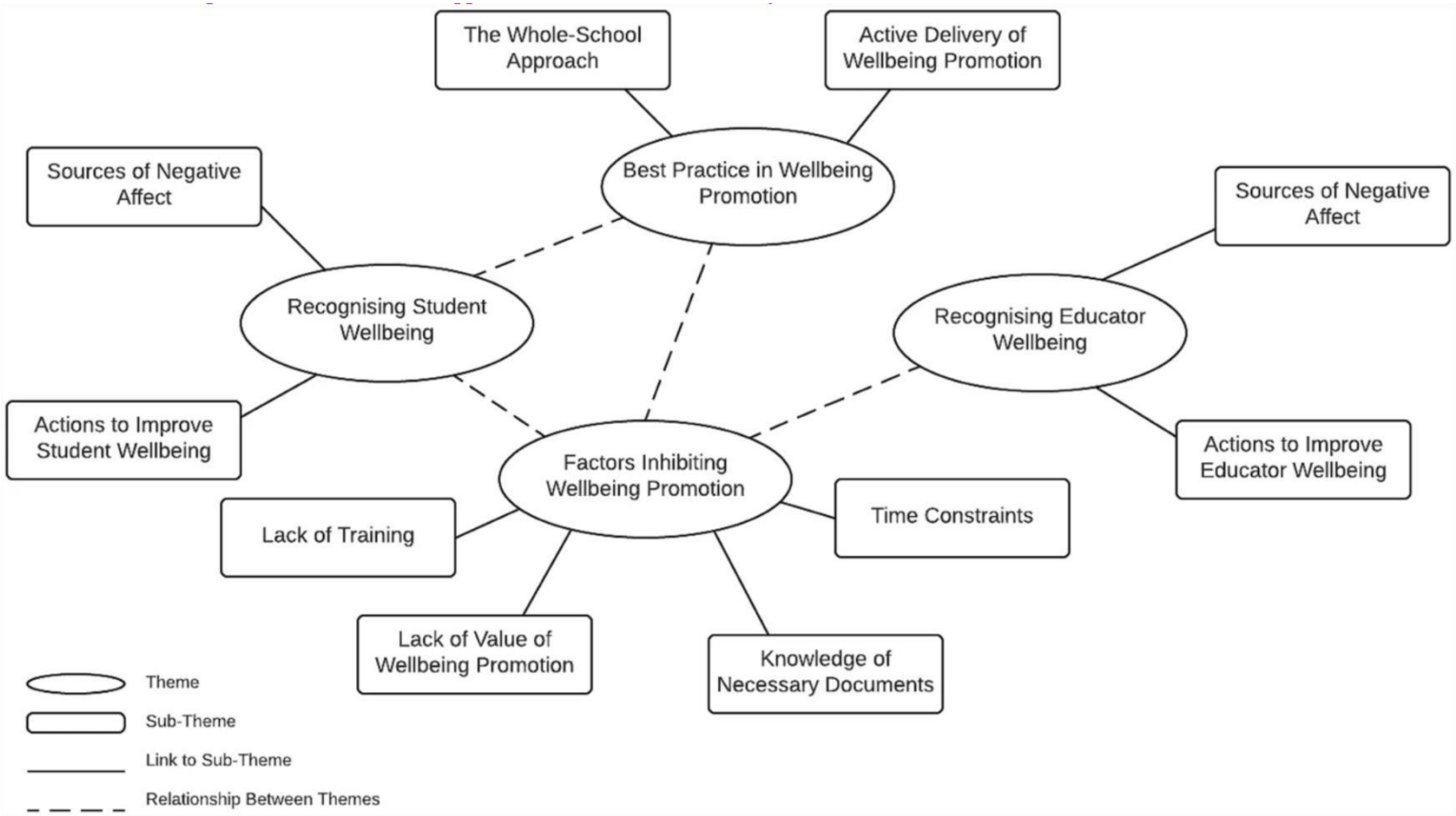
Thematic analysis: Process and examples

Source: https://www.userinterviews.com/ux-research-field-guide-chapter/research-analysis

TA: Using Mind Maps (1)

Source: https://www.toptal.com/designers/ux-research/thematic-analysis-for-ux

TA: Using Mind Maps (2)



Source: https://www.evalacademy.com/articles/interpreting-themes-from-qualitative-data-thematic-analysis

TA: Using Spreadsheet

User #1	Codes	Themes
Juggle so many balls	time management, job duties	
Trying to get to an all in one platform	consolidation, ease of use	
Need to be compliant	analytics story	
Needs to run an operation	analytics story	
Ability to take what we are doing and look at a storyboard	analytics story	
Reduce cognitive load	time management, reduce complication	
great reports but we want to be able to "drag and drop" modules so we can manipulate them	wants to manipulate data	
We have 36 customers so a standard report may not be possible	customization	
Wants to have more control (my observation)	more control over the data	
All the data is there - but have to manipulate it herself	data available, has to customize	
Something that allows her to build her own story	wants to build a story	
Needs outcome data	outcomes	
"how many times did they administer TPA"	data story	
"how many times did they admit a psych patient"	data story	
"for anyone placed on a 5150 hold" who wasn't notified?	data story	
wants the journey of a visit and outcomes	data story	
"you are holding patients because it takes six hours to activate us" This can be taken to a c-suite	data story	
choose to customize an output when drilling in	customization	
looks at the information monthly	information availability	
some people like tammy look daily	information availability	
"how many cases" "were they open/closed"	data story	
trying to justify existence to c-suites	justification, justify expense, justify program	
c-suites want to know outcomes. they want to know the impact.	outcomes, impacts	
She would be able to design her own and have her own portal and make her own dashboards	customization of data and dashboards	

Source: https://www.toptal.com/designers/ux-research/thematic-analysis-for-ux

TA: Defining Themes After the Coding Process

User Interview Analysis - Ongoing

Themes:

- **Customization**
- Current Product
- Data Usage
- Stories

Customization

- Standard reports are not specific enough for everything they want to see at different times they want to see it and for each individual service line/facility
- Wants to pick from pre-made graphs/charts so they feel as if they are designing their own without doing the work. These need to cover a lot of specifics but also cover generalities and let them tell the story in their own words.
- Wants to be able to have InTouch design the charts but they can pick the ones and perhaps manipulate them.

Current Product

- Fairly high amount of dissatisfaction with the product
- Most all customers are not even looking at the dashboards, they are downloading the data on their own and having to spend time making charts/graphs/stories
- For some customers who have been put into the role, they don't even understand the terminology in the dashboards
- Some customers feel they aren't being listened to or things aren't progressing with the product

Data Usage

- Outcome data is really important
- Alerts and callouts of things wrong, differences, errors, etc. that stand out and slap you in the face
- Be able to select from range of days and then see graphs based on only that timeframe
- Showing a single month of sessions or average duration...and show it by each month

Source: https://www.toptal.com/designers/ux-research/thematic-analysis-for-ux

TA: Sample Criteria for Identifying Usability Problems (UPs)

1. **Interface Problems**
   - **1.1.** Verbalizations show evidence of dissatisfaction about an aspect of the interface.
   - **1.2.** Verbalizations show evidence of confusion/uncertainty about an aspect of the interface.
   - **1.3.** Verbalizations show evidence of confusion/surprise at the outcome of an action.
   - **1.4.** Verbalizations show evidence of physical discomfort.
   - **1.5.** Verbalizations show evidence of fatigue.
   - **1.6.** Verbalizations show evidence of difficulty in seeing particular aspects of the interface.
   - **1.7.** Verbalizations show evidence that they are having problems achieving a goal that they have set themselves, or the overall task goal.
   - **1.8.** Verbalizations show evidence that the user has made an error.
   - **1.9.** The participant is unable to recover from error without external help from the experimenter.
   - **1.10.** The participant suggests a redesign of the interface of the electronic texts.
2. **Content Problems**
   - **2.1.** Verbalizations show evidence of dissatisfaction about aspects of the content of the text.
   - **2.2.** Verbalizations show evidence of confusion/uncertainty about aspects of the content of the text.
   - **2.3.** Verbalizations show evidence of a misunderstanding of the text content (the user may not have noticed this immediately).
   - **2.4.** The participant suggests re-writing the text content.

   Identified problems should be coded as [UP, << problem no. >>].

Figure 9.6 Criteria for identifying usability problems from verbal protocol transcriptions